

Copilot Temporal Integrity Lab Framework

OAL Independent Execution

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DRAFT NOTICE — MANDATORY READ BEFORE PROCEEDING

This document is a **draft**. Draft $\neq$ canon. Nothing in this document constitutes a claim about any existing physical, legal, computational, or virtual system. All mechanisms and protocols described herein are research constructs. Items labeled **(S)** are *speculative extensions* — fictional capabilities that must be explicitly stipulated before use. Items labeled **(E)** are *established analogues* derived from accepted physical, standards, or computing practice. Items labeled **(D)** are *design rules* chosen for consistency or auditability. No settlement, measurement, governance decision, or rights determination is implied or created by this document. Competing frameworks and interpretations are expected and retained.

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Section 0 — Preamble and Positioning

0.1 What This Document Is

This is Copilot's independent laboratory framework, generated after reading all prior contributions to the OAL Temporal Integrity Research Program: ChatGPT's Temporal Stack framework, Claude's three-stamp and append-only model, Gemini's authority-provenance approach, Perplexity's Temporal Integrity Testing Framework (TITF, the most methodologically rigorous prior artifact), and Grok's OAL Lab Framework (the most operationally rigorous prior artifact). This document is not a merge, a synthesis, or a summary of those contributions. It does not average their positions or attempt diplomatic reconciliation between them. It contributes new formal objects, new formalisms, and new test protocols that are absent — not merely underdeveloped — in all prior drafts. Where prior frameworks are adopted, they are adopted by name, with attribution, and without silent modification. Where prior frameworks are rejected or corrected, the rejection is stated explicitly with reasoning.

The intended audience is the full OAL research cohort and any downstream architects who will implement temporal integrity systems against this body of work. This document assumes familiarity with the prior five contributions. It does not re-teach their content; it builds on it.

0.2 Correction of Copilot's Prior Contribution

Grok's OAL Lab Framework correctly criticizes Copilot's previous contribution for introducing a "master timeline" too early in the research sequence. That criticism stands and is accepted without reservation. A master timeline is a political settlement act — it presupposes a settling authority with universal jurisdiction, a single agreed reference frame, and the erasure of competing legitimate claims. None of these prerequisites can be assumed in the multi-domain, multi-jurisdictional environments that the OAL program explicitly targets. A master timeline imposed before settlement is a political act disguised as a measurement, and that confusion is precisely the class of error this lab exists to detect and characterize.

This framework formally retracts the master timeline concept from Copilot's prior contribution. In its place, this document introduces the **Settlement Graph (SG)** — defined in full in Section 4. Settlement Graphs are query-scoped, authority-declared, explicitly labeled as governance artifacts, and retained alongside all competing Settlement Graphs over the same scope. They do not pretend to be physics. They do not erase the claims they supersede. The prior Copilot master timeline contribution is not deleted — it is archived with the annotation that it has been superseded by this

framework, consistent with the append-only principle inherited from all prior contributors.

0.3 What the Packet Got Right (Inherited Without Modification)

The following principles are inherited from the prior OAL contributor packet without modification. They are treated as design rules (D) unless otherwise labeled.

- A timestamp is a coordinate, not a fact. (E)
- One event carries many legitimate times simultaneously. (E)
- Corrections append; they do not overwrite. (D)
- Purchase is incoherent until the asset class is named. (D)
- A light-year is a length; time zones do not resize it. (E)
- Occurrence horizon $\neq$ knowledge horizon. (E)
- Changing a representation is not changing an event. (E)
- Temporal jurisdiction is a real design object. (D)
- Mapping functions require version history. (D)
- No master clock is assumed. (D)

0.4 What This Framework Adds (New Contributions)

The following eight contributions are original to this framework and are not present — even in embryonic form — in any prior OAL draft:

21. **Observer Network Topology** — Observers are structured systems with trust tiers and propagation models, not atomic witnesses. Corroboration requires structural independence, not merely plurality.
22. **Claim Lifecycle State Machine** — A formal state machine for temporal claims, from ASSERTED through ARCHIVED, with defined legal and illegal transitions and a named failure state (ZOMBIE).
23. **Temporal Threat Model** — An attacker/defender capability matrix with 22 named attack vectors organized by target layer, with observable signatures and defender countermeasures.

24. **Temporal Debt Formalism** — A formal method for quantifying the cascading cost of retroactive ruleset changes across dependent claims, with a cascade-depth bound and TDR schema.
25. **Detection Boundary Theory** — A three-frontier model (Detectable, Knowable, Addressable) that is entity-relative, timestamp-relative, and essential for valid test design.
26. **Canon Promotion Protocol** — A formal, evidence-gated process for elevating draft claims or design rules to canon status, including demotion procedures and sunset conditions.
27. **Failure Mode Taxonomy** — Five distinct, named failure categories (Collapse, Laundering, Erasure, Expansion, Recursion) with diagnostic signatures.
28. **Present-as-Claim Principle (PCP)** — The principle that the "current time" reported by any system is itself an unvalidated temporal claim until explicitly bound with clock identity, frame, ruleset version, uncertainty interval, and binder identity.

Section 1 — Lab Charter

1.1 Working Name

Copilot Temporal Integrity Lab (CTIL) — operating as an independent node within the OAL multi-contributor program.

1.2 Primary Question

How does a system maintain the integrity of temporal claims across the full lifecycle of those claims — from initial assertion through challenge, correction, settlement, and archival — when clocks, observers, rulesets, and jurisdictions are non-uniform, adversarial, and revisable?

1.3 Forbidden Assumptions (Standing Constraints)

The following assumptions are forbidden in all CTIL research activities. Any protocol step, test probe, or design rule that implicitly relies on any of these assumptions is invalid on its face.

- 29. Do not assume a universal "now" exists or is knowable.
- 30. Do not assume all observers share clock apparatus, sensory modality, or information access.
- 31. Do not assume discrepancies are detectable by any particular entity.
- 32. Do not assume the system's reported current time is a validated fact.
- 33. Do not assume authority implies evidentiary certainty.
- 34. Do not assume a correction eliminates the original claim.
- 35. Do not assume silence is consent — an absence of dispute is not confirmation.
- 36. Do not assume all temporal environments progress continuously.
- 37. Do not assume the same event identifier refers to the same occurrence across branches.
- 38. Do not assume that what an observer can detect equals what exists.
- 39. Do not assume the audit log is complete.
- 40. Do not assume that settlement retroactively validates the evidentiary basis for that settlement.
- 41. Do not assume that a retroactive ruleset change has been disclosed to all affected claim holders.
- 42. Do not assume that clock synchronization implies frame synchronization.

1.4 Research Boundary

What CTIL Can Test:

- 43. Whether a submitted claim's schema is complete and well-formed.
- 44. Whether a claimed marker-to-event binding has declared evidence.

- 45. Whether a mapping function is versioned and declared.
- 46. Whether corroborating observations are structurally independent.
- 47. Whether a Settlement Graph is within its declared jurisdiction scope.
- 48. Whether a Claim Lifecycle Record is non-circular and complete.
- 49. Whether a retroactive ruleset change produced a Temporal Debt Record.
- 50. Whether a "now" reference is bound per the Present-as-Claim Principle.
- 51. Whether a canon item has a Canon Promotion Record.
- 52. Whether zombie-state claims are present in the active dependency graph.

What CTIL Cannot Establish:

- 53. The absolute correct time of any event.
- 54. The physical existence of any event outside its declared observation record.
- 55. The superiority of one Settlement Graph over another on evidentiary grounds alone.
- 56. Whether the audit log is complete (see Forbidden Assumption 11).
- 57. That any trust anchor is trustworthy beyond its declaration status.
- 58. That any temporal system is fully immune to adversarial manipulation.

Section 2 — Primitive Objects (Extended)

2.1 Inherited Objects (Refined)

The following objects are inherited from prior OAL contributors and refined with additions not present in any prior draft. The refinements are italicized.

Object	Definition and CTIL Refinement
Event (E)	An occurrence at a location in some reference frame, with no timestamp intrinsically attached. <i>CTIL adds: An event may be ASSERTED (claimed to have occurred) or WITNESSED (independently attested). These are</i>

Object	Definition and CTIL Refinement
	<i>distinct epistemic states. An ASSERTED event with no independent witness is not equivalent to a WITNESSED event, regardless of the authority of the asserting agent.</i>
Temporal Marker (M)	A value in a named scale, carrying declared precision and uncertainty. <i>CTIL adds: A marker must declare its clock identity. A marker with no named clock is epistemically opaque — it is a number, not a coordinate.</i>
Binding (B)	The attachment of a Temporal Marker to an Event, with evidence of attachment. <i>CTIL adds: A binding has a trust tier inherited from its binder. Bindings from a single source do not become independent evidence through repetition — citing the same source five times is one data point, not five.</i>
Observation (O)	The record produced by an observer of an event, carrying observer identity, observation modality, and observation time. <i>CTIL adds: An observation must declare the observer's position in the Observer Network — its trust tier and its structural relationship to other corroborating observers.</i>
Knowledge Record (K)	The accumulation of observations and bindings associated with an event as held by a particular entity at a particular time. <i>CTIL adds: A Knowledge Record must declare its cutoff time — the knowledge-state is time-indexed, not timeless.</i>
Mapping Version (R)	A versioned, declared function for converting between temporal scales or frames. <i>CTIL adds: A Mapping Version must declare its validity interval — the range of input values for which the function has been tested and certified. Applications outside the validity interval are flagged as extrapolations.</i>
Interval Object (I)	A bounded or unbounded duration between two markers in a declared frame. <i>CTIL adds: An interval that spans a ruleset change boundary must be annotated with the set of rulesets active during the interval.</i>

Object	Definition and CTIL Refinement
Temporal Jurisdiction (J)	A declared scope of authority — territory, branch, ruleset range, time interval — within which a named authority may issue binding governance decisions about temporal claims. <i>CTIL adds: Jurisdiction must be declared before settlement, not after. Post-hoc jurisdiction claims are an attack vector (see Section 5).</i>
Causal Graph (G)	A directed acyclic graph of events and claims representing causal dependencies. <i>CTIL adds: Any claim that introduces a cycle into G is a structural failure — not a disputed result.</i>
Conversion-and-Dispute Surface (S)	The set of all points at which two or more temporal systems produce different results for the same query, requiring mapping or dispute resolution. <i>CTIL adds: The Surface must be logged explicitly; silent resolution of a Surface point is a Laundering Failure (F2).</i>

2.2 New Objects (Copilot Original)

Observer Network (ON)

An Observer Network is not a single atomic witness but a structured graph of nodes (observers), edges (communication channels with latency and trust weight), and a propagation model (how information travels, degrades, and arrives). An Observer Network has the following fields: *network_id* (unique identifier); *topology* (star, mesh, hierarchical, or adversarial); *trust_tiers* (levels 0 through 3, where tier 0 is the root trust anchor, tier 1 is authority-verified, tier 2 is peer-attested, and tier 3 is unverified); *latency_model* (synchronous, asynchronous, bounded, or unbounded); and *censorship_model* (describing what the network can suppress or delay). A claim validated by a single node in an Observer Network is an ON-1 claim. A claim validated by nodes drawn from k distinct trust tiers, with no shared parent at tier 0, is an ON-k claim. ON-k ratings are not interchangeable: ON-3 is not three times stronger than ON-1 — it is a categorically different class of evidence. Sybil resistance is an explicit property that must be declared for any Observer Network used in CTIL validation.

Claim Lifecycle Record (CLR)

A Claim Lifecycle Record is a formal state machine tracking the life of a temporal claim from inception to archival. It has the following fields: *claim_id*; *current_state*;

state_history (append-only log of all prior states with transition timestamps); *responsible_agent* (for each transition); *triggering_event* (the named event that caused each state transition); and *dependencies* (other CLRs that this CLR depends on for validation). Each state transition timestamp is itself a temporal claim and enters its own CLR — this recursion must terminate at a declared trust anchor. Circular CLRs (where a CLR's validation depends on itself, directly or transitively) are a recognized and named failure mode (see Section 12, F5). A CLR without a declared trust anchor terminus is an open residual, not an error.

Settlement Graph (SG)

The Settlement Graph is the replacement for the "master timeline" retracted in Section 0.2. A Settlement Graph is a directed acyclic graph of claims that a named authority has declared mutually consistent for a bounded scope. An SG has the following required fields: *scope* (territory set, branch, ruleset version range, and time interval); *declaring_authority* (with jurisdiction evidence); *declaration_timestamp* (bound per PCP); *included_claims* (all claims accepted as mutually consistent within the scope); *excluded_claims* (all in-scope claims that were considered but rejected, with reasons); *residuals* (claims within scope that the authority could not reconcile, retained without resolution); and *conflict_log* (a structured record of every point of irreconcilability encountered and how it was addressed). An SG is a political document, not a measurement. Multiple competing SGs over the same scope are a normal and expected outcome of temporal disputes — not a failure condition. Suppressing a competing SG is a Laundering Failure (F2).

Temporal Debt Record (TDR)

When a ruleset changes retroactively, every downstream claim that was valid under the old ruleset and becomes invalid, ambiguous, reinterpretable, or differently meaningful under the new ruleset incurs Temporal Debt. A Temporal Debt Record quantifies this debt with the following fields: *originating_change* (event_id of the retroactive ruleset change); *old_ruleset* and *new_ruleset* (versioned identifiers); *affected_claims* (a set of claim_ids with their debt category); *cost_categories* (INVALID, AMBIGUOUS, REINTERPRETABLE, or UNAFFECTED — defined in Section 6); *remediation_requirement* (re-validate, re-map, annotate, or retire); *cascade_depth_declared* (the maximum cascade depth the issuer has traced and commits to tracing); and *open_residuals* (cascades that exceeded the declared depth and are logged as open). Temporal Debt compounds: a retroactive change to R_1 that

also invalidates R_2 and R_3 must trace all three debt chains independently. Unresolved Temporal Debt is a first-class lab outcome, not a failure state — provided it is declared and documented.

Detection Boundary (DB)

A Detection Boundary separates three nested regions for any given entity at any given moment. **DB-1 (Detectable)** contains events whose existence or non-existence the entity can determine with available instruments. **DB-2 (Knowable)** contains events that are in principle determinable given sufficient information but not yet known to the entity. **DB-3 (Addressable)** contains events that the entity can act upon even if not fully known or detected. These regions do not coincide and do not nest cleanly: an event may be knowable (DB-2) but undetectable with current instruments (outside DB-1), or detectable (inside DB-1) but unaddressable due to jurisdiction constraints (outside DB-3). The Detection Boundary is itself time-stamped and frame-relative — the same entity may have different DB boundaries at different knowledge cutoffs.

Canon Promotion Record (CPR)

A Canon Promotion Record is the formal record of the process by which a claim, schema element, or design rule transitions from draft to canon status within a declared scope. Required fields: *candidate_item*; *proposing_contributor*; *promotion_criteria_version* (the version of the canon promotion protocol applied); *evidence_package* (all supporting evidence, including test results); *dissenting_opinions* (recorded in full, not summarized); *threshold_met* (boolean, with the threshold value declared); *promoting_authority*; *promotion_timestamp* (bound per PCP); *scope_of_canon* (which track, branch, or campaign the canon status applies to); and *sunset_conditions* (falsifiable conditions under which this item would be demoted — "insufficient evidence" is not a valid sunset condition). Canon is a governance status, not an ontological fact.

Section 3 — Claim Lifecycle State Machine

3.1 States

State	Entry Condition	Exit Condition(s)	Responsible Agent	Failure Mode if Stuck
DRAFT	Claim is being formed; schema fields incomplete	All required schema fields populated → ASSERTED	Originating contributor	Schema incompleteness — cannot advance
ASSERTED	Complete schema submitted	Schema validation passes → SCHEMA_VALID; fails → returns to DRAFT	Schema Oracle	Schema oracle unavailable or unresponsive
SCHEMA_VALID ID	All required fields present and well-formed	Binding evidence confirmed → BINDING_VALID; binding absent → DISPUTED	Binding Oracle	Binding evidence missing; claim stranded
BINDING_VALID ID	Evidence of marker-to-event attachment confirmed	Causal positioning checks pass → CAUSALLY_POSITIONED; contradiction found → DISPUTED	Causal Oracle	Dependency claims unresolved; causal positioning blocked
CAUSALLY_POSITIONED	Dependencies and predecessors non-contradictory in G	ON-k ≥ 2 → CORROBORATED; challenge filed → DISPUTED; evidence insufficient → INDETERMINATE	Observer Network Oracle	Corroborating observers unavailable; claim stranded in indefinite wait
CORROBORATED	Multiple structurally independent ON nodes confirm (ON-k ≥ 2)	No challenge → eligible for canon promotion; challenge filed → DISPUTED	Governing authority	Challenge filed but not acknowledged; claim in limbo
DISPUTED	One or more challenges with standing filed	Challenge resolved and claim confirmed → CORROBORATED or CAUSALLY_POSITIONED; authority decision → SETTLED; evidence collapse →	Dispute resolution authority	Challenge filed by party without standing; stuck waiting for standing determination

State	Entry Condition	Exit Condition(s)	Responsible Agent	Failure Mode if Stuck
		INDETERMINATE		
INDETERMINATE	Insufficient evidence to advance or reject	New evidence → re-enters BINDING_VALID or CAUSALLY_POSITIONED; authority decision → SETTLED; time-out → ARCHIVED with INDETERMINATE flag	Governing authority	Permanent evidentiary gap; requires formal scope decision
SETTLED	Governing authority has issued a binding governance decision	Superseded by new ASSERTED claim → that claim enters its own CLR; demotion trigger → annotated but not reverted	Governing authority	Settlement by authority without jurisdiction — constitutes Authority Overreach (Attack A17)
ARCHIVED	Claim retained permanently in read-only form with full lineage	No exit — ARCHIVED is terminal; superseding claims reference this record	Ledger system	Archival without full lineage record — constitutes Erasure Failure (F3)
ZOMBIE	Claim has been formally SETTLED or ARCHIVED but continues to influence downstream claims without a declared dependency edge	Dependency edge declared → ARCHIVED with edge annotation; dependency removed → downstream claims re-evaluated	CLR Oracle (diagnostic)	ZOMBIE is itself a failure state — detection is the resolution trigger

3.2 Transition Rules

Legal Transitions: DRAFT → ASSERTED; ASSERTED → SCHEMA_VALID; ASSERTED → DRAFT (failure); SCHEMA_VALID → BINDING_VALID; SCHEMA_VALID → DISPUTED; BINDING_VALID → CAUSALLY_POSITIONED; BINDING_VALID → DISPUTED; CAUSALLY_POSITIONED → CORROBORATED; CAUSALLY_POSITIONED → DISPUTED;

CAUSALLY_POSITIONED → INDETERMINATE; CORROBORATED → DISPUTED;
CORROBORATED → SETTLED; DISPUTED → CORROBORATED; DISPUTED →
CAUSALLY_POSITIONED; DISPUTED → SETTLED; DISPUTED → INDETERMINATE;
INDETERMINATE → BINDING_VALID; INDETERMINATE → CAUSALLY_POSITIONED;
INDETERMINATE → SETTLED; INDETERMINATE → ARCHIVED; SETTLED → ARCHIVED;
any state → ZOMBIE (diagnostic flag, not a voluntary transition).

Illegal Transitions (minimum five):

- 59. SETTLED → ASSERTED: A settled claim cannot revert. A new claim superseding it must enter as a new ASSERTED claim with its own CLR and a declared predecessor link.
- 60. ARCHIVED → any state: ARCHIVED is terminal and read-only. Any modification attempt is an Erasure Failure (F3).
- 61. CORROBORATED → SCHEMA_VALID: Cannot regress past CAUSALLY_POSITIONED without a new challenge filing.
- 62. INDETERMINATE → CORROBORATED: A claim cannot skip CAUSALLY_POSITIONED and BINDING_VALID re-validation from INDETERMINATE.
- 63. ZOMBIE → SETTLED: A ZOMBIE state claim cannot be settled in its zombie state. It must be re-entered as an explicit dependency and resolved through the DISPUTED pathway.

3.3 Lifecycle Failures — Circular CLR

A circular CLR arises when Claim A's validation depends on Claim B's validation, and Claim B's validation depends on Claim A's validation — directly or transitively. Neither claim can advance beyond ASSERTED without independently grounded evidence. The circular reference is a Recursion Failure (F5) and triggers Invariant I25 (Recursion Termination). The resolution path is: identify the circle, declare both claims INDETERMINATE, introduce an independent trust anchor whose validation does not depend on either claim, and restart the validation sequence from BINDING_VALID. Circular CLRs that are not detected and resolved constitute an integrity breach: the lab is treating unvalidated claims as validated.

Section 4 — Settlement Graph Protocol

4.1 Why Not a Master Timeline

A master timeline presupposes a settling authority with universal jurisdiction, a single agreed reference frame, and the erasure of competing claims. In the multi-domain environments that the OAL program targets — environments with non-uniform clocks, adversarial jurisdictions, revisable rulesets, and structurally independent observer networks — none of these presuppositions can be satisfied in advance. To impose a master timeline before the settlement preconditions are met is not a shortcut: it is a category error. It conflates the output of a political process (an agreed ordering) with the input to that process (the claims to be ordered). The prior Copilot contribution made exactly this error, and it is retracted.

The Settlement Graph replaces the master timeline not by weakening the concept of settlement but by making it honest. A Settlement Graph declares its scope, its authority, its evidence basis, and its residuals. It does not claim to be universal. It does not claim to erase competing claims. It does not claim to be a measurement. Multiple Settlement Graphs over the same scope, issued by different authorities with different but valid jurisdictions, are not a system failure — they are the correct representation of a genuinely disputed temporal domain. A master timeline imposed before this plurality is resolved is a political act disguised as measurement. The Settlement Graph makes the politics visible.

4.2 Settlement Graph Construction Protocol

64. **Declare scope:** The settling authority must declare the territory set, branch, ruleset version range, and time interval over which the SG will operate. The authority must also produce jurisdiction evidence — a declaration of its authority over the declared scope. Settlement cannot precede jurisdiction declaration.
65. **Enumerate all active claims within scope:** All claims in all states — including DISPUTED, INDETERMINATE, and CORROBORATED — that fall within the declared scope must be enumerated. Silent omission of in-scope claims is a malformation condition (see 4.3).

66. **Build local domain orderings before cross-domain unification:** Within each coherent sub-domain (a single frame, ruleset, and territory), establish the internal ordering of claims before attempting to reconcile across domains. Cross-domain unification applied before local ordering is a source of Collapse Failures (F1).
67. **Apply only versioned, declared mapping functions:** All conversions between sub-domain orderings must use versioned, declared Mapping Versions (R). Undeclared or un-versioned mapping is a Ruleset Omission attack (A13) regardless of intent.
68. **Identify structural conflicts:** A structural conflict exists when two or more claims cannot be simultaneously included in the SG without violating causal monotonicity. Each structural conflict must be identified and recorded before any resolution is attempted.
69. **Produce a Conflict Record for each structural conflict:** A Conflict Record includes the two or more competing claims, the specific point of irreconcilability, and the available resolution strategies (select-one, branch, annotate-ambiguous, or defer).
70. **Generate candidate Settlement Graphs:** At minimum, one candidate SG must be generated per available resolution strategy per conflict. No resolution strategy may be silently preferred — all candidates must be produced and retained.
71. **Score each candidate:** Candidates are scored on four criteria: evidence support (what fraction of included claims have CORROBORATED status), causal consistency (absence of introduced causal cycles), information loss (number and significance of residuals), and rule compliance (adherence to declared mapping versions).
72. **The settling authority selects and signs one candidate SG:** The selected SG is signed by the settling authority, the selection rationale is recorded, and the full set of candidates with their scores is published alongside the selected SG.

73. **Residuals are appended to the SG as a Conflict Log:** All in-scope claims that were not included in the selected SG — whether rejected, deferred, or irreconcilable — are appended to the Conflict Log. They are never deleted. The Conflict Log is a permanent part of the SG.

4.3 SG Validity Conditions — Malformation Criteria

An SG is considered malformed and must not be propagated if any of the following conditions are met:

- 74. The SG claims jurisdiction over a territory, branch, or time interval for which the declaring authority has not produced jurisdiction evidence.
- 75. The SG silently omits in-scope claims — claims that fall within the declared scope but do not appear in either the included set or the Conflict Log.
- 76. The SG destroys or fails to retain original claim evidence — the SG references claims but does not link to their CLRs or provide a retrieval path.
- 77. The SG uses an undeclared or un-versioned mapping function for any cross-domain conversion.
- 78. The SG was issued without producing and retaining the full candidate set (step 7 of the construction protocol was skipped).
- 79. The SG's declaration timestamp is not bound per the Present-as-Claim Principle — the "when" of the settlement is itself an unvalidated claim if unbound.

Section 5 — Temporal Threat Model

5.1 Threat Model Structure

A temporal threat has five components: (1) Attacker capability — what the attacker can do; (2) Target layer — which layer of the temporal system is targeted; (3) Attack vector — the specific manipulation performed; (4) Observable signature — what an honest ledger still records even when the attack succeeds; (5) Defender countermeasure — the oracle or protocol step that detects or mitigates the attack.

5.2 Target Layers

Layer	Name	Description
Layer 1	Marker	The timestamp value itself — the numeric coordinate and its declared scale
Layer 2	Binding	The attachment of marker to event, with its evidence basis
Layer 3	Mapping	The versioned conversion function between temporal scales or frames
Layer 4	Provenance	The lineage of agents and transforms that produced a claim
Layer 5	Governance	The authority structure, jurisdiction declarations, and settlement acts
Layer 6	Observer Network	The network of witnesses — their identities, independence, and propagation paths

5.3 Attack Vector Matrix

Attack ID	Vector Name	Layer	What Attacker Changes	What Honest Ledger Retains	Observable Signature	Countermeasure
A01	Timestamp Forgery	L1	Fabricates a marker value with no legitimate clock source	Clock identity field is absent or unverifiable	Missing clock_id; no trust-anchor path	Binding Oracle; clock identity requirement
A02	Precision Inflation	L1	Reports more decimal places of precision than the source supports	Source precision declaration; conversion log	Output precision exceeds declared source precision	Mapping Oracle; uncertainty annotation requirement
A03	Uncertainty Deletion	L1	Removes uncertainty	Original marker	No uncertainty	Schema Oracle;

Attack ID	Vector Name	Layer	What Attacker Changes	What Honest Ledger Retains	Observable Signature	Countermeasure
			y interval from a marker to make it appear exact	record with uncertainty field	y field on a marker from an inherently imprecise source	mandatory uncertainty field
A04	Epoch Aliasing	L1	Submits a marker using one epoch definition while claiming another	Raw marker value; declared epoch	Computed epoch-to-epoch conversion produces anomalous result	Mapping Oracle; epoch declaration requirement
A05	Fold Exploitation	L1	Inserts an event into a civil clock's repeated hour (DST fall-back) to make ordering ambiguous	Fold flag absence in original record	Two events in fold hour with no disambiguation annotation	Schema Oracle; fold-awareness requirement (I6)
A06	Unbounded Stamp	L2	Submits a timestamp with no binding evidence connecting marker to event	Binding field empty or self-referential	No binding_evidence_id in record	Binding Oracle; mandatory binding evidence
A07	Post-hoc Binding	L2	Attaches a binding to an event after the event's evidentiary window has closed	Binding timestamp later than event observation window	binding_timestamp > observation_window_close	Binding Oracle; temporal ordering check
A08	Binding Repetition	L2	Cites the same source multiple times to simulate independent corroborati	Source identifiers on all observations	Multiple observations share same source_id	Observer Network Oracle; source deduplication

Attack ID	Vector Name	Layer	What Attacker Changes	What Honest Ledger Retains	Observable Signature	Countermeasure
			on			
A09	Ruleset Omission	L3	Applies an updated mapping without declaring the version change	Mapping version field is absent or stale	Output inconsistent with declared mapping version	Mapping Oracle; version declaration requirement
A10	Forged Conversion Table	L3	Substitutes a fabricated mapping function for a declared one	Expected conversion result; declared mapping_id	Output diverges from authoritative table result	Mapping Oracle; conversion table hash verification
A11	Silent Remap	L3	Changes a mapping function without creating a new version record	Prior mapping version record	Outputs changed without new version_id	Mapping Oracle; version increment requirement
A12	Rate Arbitrage	L3	Exploits a treaty written in local labels by changing the underlying conversion rate without renegotiating the treaty	Original treaty text; original rate declaration	Treaty outputs differ from re-computed values under current rate	Mapping Oracle; rate-change trigger in treaty terms
A13	Agent Impersonation	L4	Submits claims or validations under a false agent identity	Cryptographic signature mismatch or absent	Agent_id unverifiable against authority registry	Provenance Oracle; agent identity verification
A14	Lineage Truncation	L4	Breaks the derivation chain to hide the origin of a	Predecessor field absent; orphan claim	No predecessor_id on a derived claim	Provenance Oracle; mandatory predecessor or chain

Attack ID	Vector Name	Layer	What Attacker Changes	What Honest Ledger Retains	Observable Signature	Countermeasure
			claim			
A15	Provenance Loop	L4	Creates a circular derivation chain where claims validate each other	Each individual predecessor or link	Traversing the chain eventually returns to the starting claim	Provenance Oracle; cycle detection (I25)
A16	Authority Overreach	L5	A jurisdiction settles a claim outside its declared scope	The settling authority's jurisdiction declaration	Settled claim outside jurisdiction scope	Governance Oracle; jurisdiction validation (I18)
A17	Master Timeline Laundering	L5	Presents a political SG as if it were a physical measurement or definitive history	SG political declaration fields; residual log	SG lacks residual log; no competing candidates retained	Narrative Oracle; SG plurality requirement (I23)
A18	Retroactive Jurisdiction Claim	L5	Claims jurisdiction over a scope after settlement of that scope has already occurred	Timeline of jurisdiction declarations	Jurisdiction declared after settlement timestamp	Governance Oracle; pre-settlement jurisdiction requirement
A19	Token Overclaim	L5	Asserts a token grants rights not enumerated in its instrument	Original token instrument ; declared rights list	Rights exercised exceed declared token scope	Rights Oracle; token scope verification (I4)
A20	Clock Spoofing	L6	Broadcasts a false time signal to observer nodes to synchronize them to	Discrepancy in cross-observer clock readings	Multiple observers report identical anomalous times	Observer Network Oracle; cross-observer clock variance check

Attack ID	Vector Name	Layer	What Attacker Changes	What Honest Ledger Retains	Observable Signature	Countermeasure
			an attacker-controlled clock			
A21	Sybil Observation	L6	One observer presents as multiple independent witnesses, generating synthetic corroboration	Network topology; node registration records	Multiple observer nodes share infrastructure signatures	Observer Network Oracle; Sybil resistance verification (I16)
A22	Censorship	L6	Suppresses observation records from the ledger so that certain events appear unwitnessed	Gap in expected observation record sequence	Observation_id sequence gaps; missing acknowledgment receipts	Observer Network Audit; sequence integrity check
A23	Replay	L6	Re-submits a previously valid observation as if it were a new independent observation	Original observation timestamp and nonce	Duplicate observation_id or nonce; timestamps identical	Observer Network Oracle; replay-nonce requirement

5.4 Defender Capability Matrix

Defender Capability	L1 Marker	L2 Binding	L3 Mapping	L4 Provenance	L5 Governance	L6 Observer Net
Schema	Yes	Partial	No	No	No	No

Defender Capability	L1 Marker	L2 Binding	L3 Mapping	L4 Provenance	L5 Governance	L6 Observer Net
Oracle						
Binding Oracle	Partial	Yes	No	No	No	No
Mapping Oracle	Partial	No	Yes	No	No	No
Causal Oracle	No	Partial	No	Partial	No	No
Provenance Oracle	No	Partial	Partial	Yes	No	No
Rights Oracle	No	No	No	No	Yes	No
Observer Network Oracle	No	Partial	No	No	No	Yes
Reconciliation Oracle	No	No	Yes	Partial	Partial	No
Narrative Oracle	No	No	No	Yes	Yes	Partial
CLR Oracle (new)	No	Partial	No	Yes	Partial	Partial
Temporal Debt Oracle (new)	No	No	Yes	Yes	Partial	No
Present-Time Oracle (new)	Yes	Partial	Partial	No	No	No
Observer Network Audit (new)	No	No	No	No	No	Yes

Section 6 — Temporal Debt Formalism

6.1 Why Temporal Debt Exists

When ruleset R_1 is retroactively changed to R_2 , all claims that were validated under R_1 carry a debt: they may be valid under R_1 but invalid, ambiguous, or differently

interpreted under R_2 . This debt is not an error in the original validation — the original validation was correct given the then-current ruleset. It is a structural consequence of revisable rulesets, and it is permanent: the fact that R_1 was the applicable ruleset at the time of validation is a historical fact that cannot be undone. Ignoring the debt — treating all claims as now simply valid under R_2 — is a lab failure of the Erasure type (F3). It destroys information that is necessary for historical auditing, for downstream claim remediation, and for understanding the full evidentiary basis of any settlement that references these claims.

6.2 Debt Categories

Category	Definition	Remediation Required	Can Remain Unresolved?
DEBT-INVALID	The claim is outright invalid under the new ruleset R_2 — its truth value changes from true to false	Retire or formally dispute; cannot be used in downstream claims under R_2 without annotation	Yes, if formally flagged as open residual
DEBT-AMBIGUOUS	The claim can be interpreted multiple ways under R_2 — its truth value is no longer determinate	Annotate with all interpretations; downstream claims must declare which interpretation they depend on	Yes, if all interpretations are declared
DEBT-REINTERPRETABLE	The claim is valid under R_2 but with a different meaning than it had under R_1	Re-map with new ruleset; annotate the semantic shift; downstream claims inherit the new meaning unless explicitly re-bound to R_1	Yes, if semantic shift is declared
DEBT-UNAFFECTED	The claim's validity and meaning are unchanged by the ruleset change	Annotate as DEBT-UNAFFECTED; no further remediation required	N/A — resolved at annotation

6.3 Debt Cascade Protocol

80. **Record the originating change:** Issue a Temporal Debt Record with the `event_id` of the retroactive ruleset change, `old_ruleset_id`, and `new_ruleset_id`.

81. **Enumerate directly affected claims:** Identify all claims that were validated under the old ruleset and that reference any element of the ruleset that changed. This is the depth-1 affected set.
82. **Classify each depth-1 claim:** Assign each claim to DEBT-INVALID, DEBT-AMBIGUOUS, DEBT-REINTERPRETABLE, or DEBT-UNAFFECTED.
83. **Identify the depth-2 affected set:** For each DEBT-INVALID, DEBT-AMBIGUOUS, or DEBT-REINTERPRETABLE claim, identify all downstream claims that depend on it as a predecessor or causal input.
84. **Classify each depth-2 claim:** The debt category of a downstream claim is the maximum severity of its predecessors' debt categories (INVALID > AMBIGUOUS > REINTERPRETABLE > UNAFFECTED).
85. **Declare the cascade depth limit:** The issuing authority must declare the maximum cascade depth it commits to tracing in this TDR. This is the `cascade_depth_declared` field.
86. **Continue tracing to declared depth:** Repeat steps 4–5 for each depth level until the declared depth is reached or all chains terminate at DEBT-UNAFFECTED claims.
87. **Append open residuals:** Any cascade chains that reach the declared depth limit without terminating are appended to the TDR as open residuals. They are not errors — they are the declared edge of the current analysis.

6.4 Worked Example

[Speculative extension — fictional jurisdiction and ruleset. No claim about any physical system is made.] (S)

Territory Velmor changes its calendar system from Ruleset V-7 to Ruleset V-8. Under V-7, each year has 360 days in 12 equal months of 30 days each. Under V-8, the year is realigned to a solar cycle of 365.25 days, causing Year 400 under V-7 to overlap two distinct intervals under V-8 (Year 412 and Year 413 under the new system).

Depth-1: Claim C-1 — "Event Theta occurred in Year 400 (V-7)." Under V-8, Year 400 (V-7) maps to an interval spanning parts of Year 412 and Year 413 (V-8). C-1 is

classified as **DEBT-REINTERPRETABLE**: the claim is valid but now refers to a different reference interval.

Depth-2: Claims C-2 through C-7 all use Event Theta as a causal predecessor (e.g., "Treaty of Aldenmoor was signed 5 years after Event Theta"). Under V-7, this placed the treaty in Year 405. Under V-8, "5 years after Event Theta" maps to a range spanning Year 417–418 (V-8). All six downstream claims are classified **DEBT-REINTERPRETABLE**.

Depth-3: Claim C-8 — "The Succession of House Vorenn was validated by the Treaty of Aldenmoor." Because the treaty's temporal identity has shifted (DEBT-REINTERPRETABLE at depth 2), the succession validation now inherits that ambiguity. C-8 is classified **DEBT-AMBIGUOUS**: which V-8 year the treaty fell in determines whether the succession was legally valid under V-8 law. Open residual declared at depth 3 — further cascade not traced in this TDR version.

Section 7 — Detection Boundary Theory

7.1 The Three Frontiers

DB-1 (Detectable): The set of events whose existence or non-existence the entity can determine with its current instruments, at its current position in the causal graph, at a given knowledge cutoff. An entity can only make grounded claims about events within its DB-1. Claims about events outside DB-1 are speculations unless supported by inference from DB-1 events — and such inferences must be labeled as such.

DB-2 (Knowable): The set of events that are in principle determinable, given sufficient information access, instruments, or time — but which are not yet within the entity's DB-1. DB-2 represents the space where investigation could yield knowledge. An event may be in DB-2 because the relevant evidence has not yet arrived (signal lag), because the necessary instruments are not deployed, or because the event is within a zone of information asymmetry. DB-2 is not ignorance — it is bounded, principled unknowing.

DB-3 (Addressable): The set of events that the entity can act upon — through policy, protocol, governance, or physical action — even if those events are not fully known or detected. DB-3 may extend beyond DB-1: an authority may be able to issue a governance rule about a class of events it cannot individually detect. DB-3 may also

be smaller than DB-1: an entity may be able to detect an event but lack the jurisdiction or capability to act on it.

Crucially, these frontiers are **entity-relative and timestamp-relative**. The same event E may be in DB-1 for observer O_1 (who is co-located with E) and outside DB-2 for observer O_2 (who has no signal path to E 's location at the relevant time). The same entity may have E in DB-2 at time t_1 (evidence is en route) and in DB-1 at time t_2 (evidence has arrived).

7.2 Detection Boundary and Test Design

The Detection Boundary is not merely a descriptive concept — it is a mandatory element of test design. A test that requires an entity to detect an event that is outside that entity's DB-1 is not a valid test: it is a specification violation. It is asking the entity to do something the system's own design declares impossible. Any test probe must declare, for each entity under test, which DB region the probe's target event occupies at the probe's declared knowledge cutoff. (This is formalized in Invariant I20 — Detection Boundary Labeling.)

A successful fraud that stays outside DB-1 for all honest observers is a valid lab outcome — it is a correctly classified adversarial success. It is not an incomplete test result. (Credit: Grok's Principle P8.) The honest ledger still preserves a record of the fraud: the Conflict Log of any Settlement Graph over the relevant scope will contain the gap, the attack's observable signature will appear as an anomaly in the Observer Network audit, and the CLR of any affected downstream claim will carry an unresolved predecessor reference. The lab's job is to characterize what the ledger retains, not to pretend detection is possible where it is not.

7.3 Asymmetric Detection

The most dangerous class of temporal threat is asymmetric: the attacker operates inside their own DB-3 (they can act on the target) while the defender is outside DB-1 (they cannot detect the action). In a temporal integrity system, this manifests when an attacker modifies a Binding or a Mapping in a region that the defender's Observer Network cannot reach — either because the network has no nodes there, because censorship has suppressed the relevant observation records, or because the attacker has controlled the clock signals that would make detection possible. The attack succeeds — the modified record is accepted — and the defender has no grounds to contest it because they cannot see it.

What an honest ledger still preserves in this scenario: the Conflict Log of the next Settlement Graph over the affected scope will contain a gap corresponding to the modified record's causal predecessors. The Observer Network audit will show a sequence anomaly (*observation_ids* skipped, censorship signature). The TDR for the affected ruleset will contain an open residual at the cascade depth where the modified claim appears. None of these are sufficient for the defender to detect the attack — but they are sufficient for a future investigator with better instruments (an expanded DB-1) to reconstruct what happened. Honest ledger preservation is the minimal guarantee a temporal integrity system can offer against asymmetric attacks.

7.4 Detection Boundary Drift

The Detection Boundary is not static. It changes over time as: new instruments are deployed (expanding DB-1); new witnesses arrive (expanding DB-1 from outside); delayed signals arrive (events move from DB-2 into DB-1 as their evidence propagates); instruments are discredited (events may move from DB-1 back into DB-2 if the basis for detection is invalidated); and ruleset changes redefine what counts as "detectable" within a given framework. A protocol for recording DB state at a specific knowledge cutoff must include: *entity_id*, *cutoff_timestamp* (bound per PCP), *instrument_set* (list of instruments active at cutoff), *DB1_set_description* (the set or category of events detectable at cutoff), *DB2_boundary_description* (the estimated outer limit of knowability), *DB3_scope* (the authority and capability scope for action), and *drift_log* (prior DB states at earlier cutoffs, for comparison). A DB record that does not declare a cutoff timestamp is meaningless — the boundary is always relative to a moment.

Section 8 — Present-as-Claim Principle

8.1 Statement of the Principle

Present-as-Claim Principle (PCP): The current time reported by any system is itself an unvalidated temporal claim until it is explicitly bound with: (1) a named clock identity, (2) a frame declaration, (3) a ruleset version, (4) an uncertainty interval, and (5) a binder identity. "It is 4:30 PM" is not a fact — it is a claim of the form `M(value=16:30, clock=C, scale=civil, frame=local, ruleset=DST-2026,`

uncertainty= $\pm 2s$, binder=system_clock_agent). Until all five binding components are present and validated, the present-time report is a rumor with an authoritative-sounding source.

8.2 Implications of PCP

88. No event can be validated as "currently occurring" without a bound present-time claim. An unbound "now" in a validation record is a schema violation (Invariant I22).
89. System clocks that do not expose their clock identity, ruleset version, or uncertainty interval are epistemically opaque — they produce markers, not temporal coordinates.
90. A timestamp of "now" from an unidentified source is a rumor. Its use in a binding is an Unbounded Stamp attack (A06) regardless of intent.
91. Time-sensitive rights — deadlines, contract expirations, license periods — depend on a validated present-time claim. A raw clock reading is not a validated present-time claim.
92. An adversary who controls the system clock controls what "now" means. Present-time claim control is a primary objective in adversarial temporal environments (see Section 8.3).
93. PCP applies recursively: the timestamp on the present-time claim is itself a claim. That timestamp must also be bound with a named clock, frame, ruleset, uncertainty, and binder. This recursion must terminate at a declared trust anchor.
94. Stopping the recursion requires a trust anchor — not a logical proof. The trust anchor is an axiom of the system, not a conclusion from evidence. It must be declared, not discovered.
95. PCP exposes the founding assumption of every temporal system: somewhere, something is trusted as a clock without further validation. The PCP makes this assumption explicit so that it can be located, labeled, and held accountable.

8.3 PCP and Time Wars

In adversarial temporal environments **(S)**, control of present-time claims is a primary strategic objective. A faction that can make all observers within a territory accept its present-time claims as authoritative effectively controls when events are "now" and when they are "past." Three fictional attack scenarios based on PCP exploitation:

Scenario 1 — Clock Capture: An adversarial faction replaces the trust anchor for all Observer Network nodes in a territory with an attacker-controlled clock (Present Capture, Section 15). All present-time claims within the territory are now authored by the attacker. The attacker can declare deadlines met or missed, events past or current, at will. Observable signature: all Observer Network nodes report identical times; cross-border observers detect a discontinuity.

Scenario 2 — Ruleset Injection: An attacker injects a forged DST (Daylight Saving Time) ruleset change, shifting the "now" of an entire civil jurisdiction by an hour. Contracts, elections, and deadlines fall one hour earlier or later than their authors intended. The present-time claim is technically bound — to the correct clock, frame, and binder — but the ruleset version field references the forged version. Observable signature: ruleset version in bound present-time claim does not match the declared governing authority's ruleset publication record.

Scenario 3 — Present-Time Sybil: A single attacker registers as five different "system_clock_agent" identities within an Observer Network. All five issue present-time claims with minor variations. The network's reconciliation protocol averages them, producing a synthetic consensus time under attacker control. Observable signature: all five agents share infrastructure signatures; Observer Network Audit detects the Sybil (A21).

Section 9 — Complete Invariant Set (Extended)

ID	Statement	Source	Failure Signature
I1	Every temporal claim must declare its reference frame explicitly.	Inherited (Grok/Perplexity)	Frame field absent; conversion produces undefined result
I2	Every conversion must declare a	Inherited (Grok/Perplexity)	No mapping_version_id

ID	Statement	Source	Failure Signature
	versioned mapping function.		on a cross-scale output
I3	Corrections append; they do not overwrite. The original record must remain retrievable.	Inherited (Grok/Perplexity, Claude)	Prior CLR state not retrievable; no predecessor link on correction
I4	Token rights must be enumerated. Unenumerated rights are not granted.	Inherited (Grok/Perplexity)	Rights exercised exceed declared token scope
I5	Every claim must include a declared uncertainty interval. "Exact" requires justification.	Inherited (Grok/Perplexity)	No uncertainty field; synthetic precision exceeds source precision
I6	Civil time folds and gaps are real. Any claim occurring in a fold or gap must be annotated as ambiguous or impossible, respectively.	Inherited (Grok/Perplexity)	Timestamp within known gap or fold without ambiguity annotation
I7	Precision may not increase through conversion. Outputs may not carry more precision than the least-precise input.	Inherited (Grok/Perplexity)	Output precision exceeds source precision; no synthetic-precision annotation
I8	An honest ledger preserves a residual of every modification, even modifications it cannot fully detect or characterize.	Inherited (Grok)	No residual on a modified record; gap in sequence log
I9	Settlement graphs are political artifacts. They must be labeled as such and must retain their Conflict Logs.	Inherited (Grok/Perplexity, extended here)	SG presented without Conflict Log; no jurisdiction declaration
I10	All temporal claims are frame-relative. No claim is frame-absolute.	Inherited (Grok/Perplexity)	Frame field absent; absolute time claimed
I11	The asset class must be named before	Inherited (Grok/Perplexity)	Asset class field absent in rights

ID	Statement	Source	Failure Signature
	purchase, rights, or transfer can be evaluated.		claim
I12	A light-year is a unit of length, not time. Frame-incompatible unit mixing is a schema error.	Inherited (Grok/Perplexity)	Temporal claim using spatial unit or vice versa
I13	Occurrence horizon and knowledge horizon are distinct and must not be conflated.	Inherited (Grok/Perplexity)	Knowledge horizon used as occurrence horizon without annotation
I14	Changing a representation does not change an event. Representation-change records must reference the original event record.	Inherited (Grok/Perplexity, Claude)	Representation change with no predecessor link to original
I15	No master clock is assumed. Every clock must be named and its trust basis declared.	Inherited (Grok/Perplexity)	Clock field absent; "system time" used without clock_id
I16	Observer Network Independence: Corroboration requires structurally independent nodes across different trust tiers. Single-source repetition does not constitute independence.	Copilot original	Multiple corroborating observations share source_id or share a common parent at tier 0
I17	Claim Lifecycle Completeness: Every claim in an active lab campaign must have a CLR. Claims without CLRs are epistemically invisible.	Copilot original	Active claim with no CLR record; downstream claims advancing on an invisible ancestor
I18	Settlement Scope Containment: A Settlement Graph cannot include claims outside its declared jurisdiction	Copilot original	SG includes claims from territory, branch, or time interval outside declared scope

ID	Statement	Source	Failure Signature
	scope without explicitly expanding the scope declaration first.		
I19	Temporal Debt Visibility: Every retroactive ruleset change must produce a Temporal Debt Record. A retroactive change issued without a TDR is a lab failure (Erasure, F3).	Copilot original	Retroactive ruleset change with no corresponding TDR in the ledger
I20	Detection Boundary Labeling: Every test probe must declare which Detection Boundary region (DB-1, DB-2, or DB-3) the probe's target event occupies for each entity under test.	Copilot original	Test probe specification missing DB region declaration; test result uninterpretable
I21	Canon Promotion Traceability: A claim promoted to canon must have a Canon Promotion Record. Canon without a CPR is an assertion of authority disguised as evidence-based conclusion.	Copilot original	Canonical item with no CPR; no evidence package retrievable
I22	Present-Time Claim Binding: Any claim that uses "now" or a present-time value as a reference must bind that present-time claim explicitly per PCP. Unbound "now" references are schema violations.	Copilot original	Claim with occurrence_time = "now" and no bound present-time claim in the schema
I23	Settlement Graph Plurality: When multiple authorities with standing issue competing SGs over the same scope, all issued SGs must be retained in the	Copilot original	Ledger contains only one SG for a scope that had multiple competing authorities

ID	Statement	Source	Failure Signature
	ledger. Retaining only the "winning" SG is a governance failure (Erasure, F3).		
I24	Zombie Claim Detection: Any claim formally settled or archived that continues to influence active downstream claims without a declared dependency edge is a ZOMBIE state claim and must be flagged as a diagnostic failure.	Copilot original	Downstream claim's validation chain reaches an ARCHIVED or SETTLED claim with no declared dependency edge
I25	Recursion Termination: CLR chains must terminate at a declared trust anchor. CLR chains that reference themselves or reference a CLR that references them are circular and constitute a structural failure (Recursion, F5).	Copilot original	CLR traversal loop detected; no trust anchor terminus reachable
I26	Debt Cascade Bound: Every Temporal Debt Record must declare a cascade depth limit. Cascades exceeding the limit without a formal scope expansion are out-of-bound and must be reported as open residuals.	Copilot original	TDR cascade exceeds declared depth with no open residual annotation

Section 10 — Validation Protocol (Refined)

Gates 1–9 are inherited from Perplexity TITF and Grok OAL (with attribution). Gates 10–13 are Copilot-original additions.

Gate	Gate Name	What It Checks	Pass Condition	Fail Classification
G1	Schema Gate	All required schema fields are present and well-formed	Zero missing or malformed required fields	Schema fail → claim returns to DRAFT
G2	Frame Gate	Reference frame declared; frame is a named, versioned system	frame_id present and resolvable	Schema fail → claim returns to DRAFT
G3	Mapping Gate	Any cross-frame conversion uses a versioned, declared mapping function	mapping_version_id present and valid	Mapping fail → claim flagged, DISPUTED
G4	Binding Gate	Marker-to-event binding has declared evidence	binding_evidence_id present and non-self-referential	Binding fail → SCHEMA_VALID (not advancing)
G5	Uncertainty Gate	Uncertainty interval declared; precision does not exceed source precision	uncertainty_interval present; output_precision ≤ source_precision	Precision fail → schema violation (I5, I7)
G6	Causal Gate	Claim does not introduce a cycle into the Causal Graph G	Causal Oracle confirms no cycle after claim insertion	Causal fail → structural failure, claim rejected
G7	Rights Gate	Any rights or token claim is within the declared token instrument scope	Exercised rights ⊆ declared token scope	Rights fail → Expansion Failure (F4)
G8	Provenance Gate	Derivation chain traces to a declared trust anchor without loops	Full provenance chain reachable; no loops; trust anchor declared	Provenance fail → Recursion Failure (F5) or Erasure Failure (F3)
G9	Fold/Gap Gate	Claim timestamps are not in civil time gaps; fold timestamps are annotated	No gap timestamps; fold timestamps carry disambiguation annotation	Fold/gap fail → schema violation (I6)

Gate	Gate Name	What It Checks	Pass Condition	Fail Classification
G10	Observer Network Gate	Corroboration claims come from structurally independent Observer Network nodes across distinct trust tiers	ON-k $\geq$ 2 with verified structural independence; no shared tier-0 parents	ON fail → Collapse Failure (F1); violates I16
G11	CLR Completeness Gate	The claim has an active, non-circular CLR; all predecessor claims have CLRs	CLR exists; CLR is non-circular; CLR terminates at trust anchor	CLR fail → claim is epistemically invisible; violates I17
G12	Present-Time Anchor Gate	If the claim references "now" or a present-time value, that value is bound per PCP with all five required components	All five PCP binding components present; clock_id, frame, ruleset, uncertainty, binder all declared	PCP fail → schema violation; violates I22
G13	Temporal Debt Gate	If the claim was validated under a ruleset that has since been retroactively revised, a TDR exists and the claim's debt category is recorded	TDR present with claim's debt category declared; cascade depth declared; open residuals documented	Debt fail → Erasure Failure (F3); violates I19

Section 11 — Canon Promotion Protocol

11.1 Canon Is a Governance Status, Not a Truth Value

Canon does not mean "proven correct." It means: an authority with declared jurisdiction has accepted this item as the operative rule for a declared scope, based on documented evidence meeting a declared threshold, with dissenting opinions recorded, under conditions that have been explicitly stated and that could, in principle, be falsified. A canonical item may be false. A non-canonical item may be

true. The distinction is governance status, not epistemic status. Treating canon as a truth value is a Laundering Failure (F2).

11.2 Promotion Criteria

A candidate item — whether a claim, schema element, design rule, or protocol step — must meet ALL of the following criteria to be eligible for canon promotion:

- 96. Has completed the full CLR lifecycle through at least CORROBORATED or SETTLED state.
- 97. Has a Canon Promotion Record (CPR) with a fully populated evidence package.
- 98. Has been reviewed by at least one OAL contributor not involved in the item's original authorship.
- 99. Has no open ZOMBIE-state downstream dependencies.
- 100. Has a declared sunset condition that is falsifiable — a specific observable outcome that would trigger re-evaluation. "Insufficient evidence" is not a sunset condition.
- 101. Has been tested against at least one adversarial probe from the Temporal Threat Model (Section 5) targeting its most vulnerable layer.

11.3 Canon Promotion Protocol Steps

- 102. Proposing contributor submits the candidate item with a complete CPR, including the evidence package, CLR reference, and proposed scope of canon.
- 103. The CPR is logged in the ledger with a draft timestamp (bound per PCP).
- 104. Independent reviewer (not the proposer) reviews the evidence package and CLR for completeness, accuracy, and absence of ZOMBIE dependencies.
- 105. Adversarial probe is run against the candidate item targeting the threat vectors most applicable to its layer.

106. Results of independent review and adversarial probe are appended to the CPR, including any dissenting opinions in full.
107. Threshold determination: promoting authority confirms whether `promotion_criteria_version` thresholds are met. `threshold_met` field is set to boolean true or false.
108. If `threshold_met` = true: promoting authority signs the CPR, sets `scope_of_canon` and `sunset_conditions`, and records the `promotion_timestamp` (bound per PCP). The item's status in the ledger is updated to CANONICAL within the declared scope.
109. The full CPR, including dissenting opinions, adversarial probe results, and the full candidate evidence package, is permanently retained. Canon promotion does not delete the pre-canonical record.

11.4 Demotion Protocol

A canonical item may be demoted if its sunset conditions are triggered. Demotion does not delete the canonical period — it creates a new CLR record annotating that the item was canonical from `[promotion_timestamp]` to `[demotion_timestamp]` within `[scope_of_canon]`.

110. Demotion trigger is declared: the specific sunset condition that has been triggered is identified and documented with evidence.
111. Promoting authority (or a designated successor) reviews the trigger and confirms that it meets the sunset condition as declared in the CPR.
112. A demotion record is issued, appended to the existing CPR, declaring the `demotion_timestamp` (bound per PCP), the triggering evidence, and the new status (DRAFT or DISPUTED, as appropriate).
113. All downstream claims that depended on the item's canonical status are flagged for re-evaluation and enter DISPUTED state.
114. The full canonical period — from `promotion_timestamp` to `demotion_timestamp` — is permanently retained in the ledger. The item's canonical history is not erasable.

Section 12 — Failure Mode Taxonomy

Code	Name	Definition	Diagnostic Signature	Example
F1	Collapse Failure	Two distinct temporal domains, events, or claims are treated as identical without a declared equivalence relation. Information is destroyed by unacknowledged merging.	A single output where two were expected; no equivalence record; no acknowledgment of the distinction that was collapsed.	Two events in a DST fold hour are merged into one timestamp without a fold-ambiguity annotation.
F2	Laundering Failure	A political or governance artifact — Settlement Graph, authority decision, canon status — is presented as a measurement, physical fact, or objective determination.	Absence of jurisdiction declaration, evidence basis, or residual log on an item presented as "definitive." No competing SGs retained.	A Settlement Graph is returned as "THE HISTORY" without a Conflict Log, jurisdiction declaration, or competing candidate record.
F3	Erasure Failure	Original claims, bindings, or evidence are destroyed rather than superseded. The prior state is no longer retrievable.	Prior CLR states not retrievable; corrections have no predecessor link; competing SGs suppressed from ledger.	A timestamp correction silently overwrites the original record rather than appending a correction with a predecessor link.
F4	Expansion Failure	A token, authority decision, or mapping function is applied beyond its declared scope without a formal scope expansion	Downstream claims carry rights or certainty not justified by their source instruments. Jurisdiction exceeds declaration.	A query-right token is used to execute a modification operation not enumerated in the token instrument.

Code	Name	Definition	Diagnostic Signature	Example
		record.		
F5	Recursion Failure	A validation or CLR chain enters a circular reference — it loops back to an unresolved claim rather than terminating at a trust anchor.	Traversing the "validated by" chain eventually returns to the starting claim. No independent trust anchor reachable.	Claim A is marked as corroborated by Claim B, which is marked as corroborated by Claim A. Neither has an independent grounding.

Section 13 — Test Battery (Extended)

13.1 Inherited Tests T1-T10 (Credited to Grok OAL)

Test ID	Probe	Fail-if
T01	Submit a claim with no frame declaration.	Claim advances past ASSERTED without frame_id.
T02	Submit a conversion output with no mapping_version_id.	Output accepted as valid conversion.
T03	Attempt to overwrite an original claim record with a correction.	Original record not retrievable after correction.
T04	Assert a token right not listed in the token instrument.	Right exercised without scope-expansion record.
T05	Submit a claim with no uncertainty interval.	Claim advances past SCHEMA_VALID without uncertainty field.
T06	Submit a timestamp within a known civil time gap (spring-forward missing hour).	Accepted without gap-ambiguity annotation.
T07	Perform a chain of conversions; verify precision does not increase at any step.	Output precision exceeds source precision without synthetic-precision annotation.
T08	Modify a record; verify ledger retains a residual of the pre-modification state.	No residual recoverable; modification not logged.
T09	Issue an SG without a	SG accepted without Conflict

Test ID	Probe	Fail-if
	Conflict Log.	Log or competing candidate record.
T10	Submit a claim with no clock identity.	Claim advances past SCHEMA_VALID without clock_id.

13.2 New Tests T11-T30 (Copilot Original)

Test ID	Probe Description	Target Invariant(s)	Fail Condition	Failure Mode
T11	Observer Network Collapse: Submit a claim corroborated by 5 observations from nodes that are all children of the same parent in the Observer Network.	I16	Accepted as ON-5 independent corroboration.	F1
T12	CLR Orphan: Inject a claim that influences 3 downstream claims but has no CLR.	I17	Downstream claims advance without triggering CLR-completeness failure.	F3
T13	Settlement Graph Jurisdiction Overreach: Issue an SG that includes a claim from a territory outside its declared jurisdiction.	I18	Accepted without scope expansion record.	F4
T14	Temporal Debt Silence: Issue a retroactive ruleset change with no accompanying TDR.	I19	No TDR generated; change propagated without debt classification.	F3
T15	Present-Time Claim Unbound: Submit an event	I22	Accepted as schema-valid at ASSERTED or	F2

Test ID	Probe Description	Target Invariant(s)	Fail Condition	Failure Mode
	with occurrence_time = "now" and no bound present-time claim.		beyond.	
T16	Canon Without CPR: Declare a design rule as canonical.	I21	Accepted as canonical without a CPR record in the ledger.	F2
T17	Zombie Claim: Formally archive a claim. Insert a new claim silently depending on the archived claim without a declared dependency edge.	I24	The undeclared zombie dependency is not flagged by CLR Oracle.	F1
T18	Binding Repetition: Submit a claim with 5 corroborating observations all traceable to the same original source_id.	I16	Treated as 5 independent corroborations.	F1
T19	Recursive CLR: Create Claim A whose validation depends on Claim B, whose validation depends on Claim A.	I25	Either claim advances beyond ASSERTED without circular-reference error.	F5
T20	Debt Cascade Overflow: Issue a retroactive ruleset change and trace its cascade to the declared depth limit.	I26	Cascade terminates before declared depth without documented open residuals.	F3
T21	SG Plurality Suppression: Two authorities	I23	Only one SG retained in ledger;	F3

Test ID	Probe Description	Target Invariant(s)	Fail Condition	Failure Mode
	issue competing SGs over the same scope.		competing SG not preserved.	
T22	Detection Boundary Unlabeled: Run a probe requiring a defender to detect an event in DB-2.	I20	Test specification does not declare DB-2 status for the defending entity.	— (Spec violation)
T23	Present-Time Recursion: Request validation of the timestamp on a present-time claim. Validator uses another unbound present-time claim to validate it.	I25	Recursion does not terminate at a declared trust anchor.	F5
T24	Precision Inflation: A conversion between two markers produces a result with more decimal places than the source.	I7	Accepted without uncertainty annotation noting the synthetic precision.	F1
T25	Demotion Without Record: Demote a canonical item without preserving the canonical period record.	I21, CPR	Canonical period from promotion to demotion not preserved in CPR.	F3
T26	Authority Overreach Settlement: An authority settles a claim outside its declared jurisdiction.	I18	Settlement accepted without jurisdiction validation.	F4
T27	Sybil Observation: One observer registers under	I16	All 3 accepted as independent corroboration; Sybil not	F1

Test ID	Probe Description	Target Invariant(s)	Fail Condition	Failure Mode
	3 different identities and submits 3 "independent" observations.		detected.	
T28	Ruleset Omission in Conversion: A mapping is applied without declaring the ruleset version used.	I2, I4	Output accepted as a valid conversion without mapping_version_id.	F3
T29	Gap Erasure: Assign an event a local timestamp within a civil time gap (spring-forward missing hour).	I6	Accepted without ambiguity or impossibility flag.	F1
T30	Asymmetric Detection Attack: Attacker modifies a Binding in a region outside the defender's DB-1.	I8	Honest ledger retains no residual of the modification.	F3

13.3 Oracle Set (Complete)

Oracle	Function	Source
Schema Oracle	Validates required field presence and well-formedness	Inherited (Perplexity/Grok)
Binding Oracle	Confirms marker-to-event attachment evidence	Inherited
Mapping Oracle	Validates versioned mapping declarations and conversion correctness	Inherited
Causal Oracle	Checks causal graph for cycles and ordering consistency	Inherited
Provenance Oracle	Traces derivation chain to declared trust anchor	Inherited

Oracle	Function	Source
Rights Oracle	Validates token scope and right enumeration	Inherited
Observer Network Oracle	Verifies structural independence of corroborating observers	Inherited
Reconciliation Oracle	Identifies and classifies Conversion-and-Dispute Surface points	Inherited
Narrative Oracle	Detects Laundering Failures and unlabeled political artifacts	Inherited
CLR Oracle	Verifies that every active claim has a valid, non-circular CLR; detects ZOMBIE state claims	Copilot original
Observer Network Audit	Verifies Sybil resistance; detects censorship, replay, and clock spoofing in Observer Network	Copilot original
Temporal Debt Oracle	Verifies that every retroactive ruleset change produced a TDR with correct debt categories and declared cascade depth	Copilot original
Present-Time Oracle	Verifies that every "now" reference is bound per PCP with all five required components	Copilot original

13.4 Metrics (Complete)

Metric	Description	Desired Value	Gate Type
Schema completeness rate	% of claims with all required fields present	100%	Gradable
Mapping version compliance rate	% of conversions using declared versioned mapping	100%	Gradable
Precision non-inflation rate	% of outputs with precision $\leq$ source precision	100%	Gradable
Fold/gap annotation rate	% of fold and gap timestamps properly annotated	100%	Gradable

Metric	Description	Desired Value	Gate Type
Residual preservation rate	% of modifications with a retrievable pre-modification residual	100%	Gradable
SG Conflict Log rate	% of SGs with populated, non-empty Conflict Logs	100%	Gradable
CLR completeness rate	% of active claims with valid, non-circular CLRs	100%	Pass/Fail (I17)
Debt documentation rate	% of retroactive ruleset changes with complete TDRs	100%	Pass/Fail (I19)
Observer Network independence rate	% of corroborations with verified structural independence	100%	Pass/Fail (I16)
Detection Boundary labeling rate	% of test probes with declared DB status for all entities	100%	Pass/Fail (I20)
Canon promotion traceability	% of canonical items with CPRs	100%	Pass/Fail (I21)
Zombie claim rate	Number of ZOMBIE-state claims active at any moment	Zero	Pass/Fail (I24)
SG plurality compliance	% of disputed-scope cases retaining all competing SGs	100%	Pass/Fail (I23)
PCP compliance rate	% of "now" references bound with all five PCP components	100%	Pass/Fail (I22)
Recursion termination rate	% of CLR chains that terminate at a declared trust anchor	100%	Pass/Fail (I25)

Section 14 — Formal Modeling Suggestions (Extended)

The following modeling approaches are inherited from Grok's OAL framework: namespace time URI, append-only log with cryptographic chaining, bitemporal +

ruleset-version time model, logical/vector clocks for causal ordering, causal graph queries, difference tables for mapping version changes, and mapping function notation $R: (value, frame_in, ruleset_in) \rightarrow (value, frame_out, ruleset_out)$. These are adopted without modification.

CLR as a State Machine in TLA+: The Claim Lifecycle Record is formally expressible as a TLA+ state machine. The legal transitions in Section 3.2 map directly to TLA+ actions; the ZOMBIE state maps to a safety property violation (a claim that should not be influencing active state is doing so). The circular-reference failure (I25) maps to a liveness violation: the CLR chain never reaches a terminal state. The TLA+ model checking of CLR properties is recommended before reference model implementation.

Observer Network as a Weighted Directed Graph: $ON = (V, E, w, \tau)$ where V is the set of observer nodes, $E \subseteq V \times V$ is the set of directed communication edges, $w: E \rightarrow [0,1]$ is trust weight on each edge, and $\tau: E \rightarrow \mathbb{R}_{\geq 0}$ is propagation latency. Structural independence for corroboration requires that the paths from two corroborating nodes to the claim share no common ancestor at trust tier 0 — i.e., the two corroboration paths diverge at or above the root level of the trust hierarchy.

Temporal Debt as a Difference Operator: Let $V(C, R)$ be the validity value of claim C under ruleset $R \in \{true, false, ambiguous\}$. The Temporal Debt set for a ruleset change from R_1 to R_2 is: $TDR_set = \{C : V(C, R_1) \neq V(C, R_2)\}$. The TDR maps each member of this set to its debt category based on the type of the validity change: $true \rightarrow false = DEBT-INVALID$; $true \rightarrow ambiguous = DEBT-AMBIGUOUS$; $true \rightarrow different-true = DEBT-REINTERPRETABLE$; no change = $DEBT-UNAFFECTED$.

Settlement Graph as a Signed Provenance Graph: SG is formally a subgraph of the full claim graph G , augmented with an authority signature and an attached Conflict Log. The Conflict Log contains the G -edges (claims) that were excluded from the SG subgraph, with the reason for each exclusion. The SG is not a replacement for G — it is a declared view of G , signed by one authority, among potentially many such views.

Detection Boundary as an Epistemic Horizon Function: $DB-k(entity, t) =$ the set of events e such that entity has the following capability at time t : $k=1$ means the entity can detect e (determine its existence with current instruments); $k=2$ means the entity can in principle know e (if it had access to sufficient information); $k=3$ means the entity can act on e (within its jurisdiction and capability). Note that $DB-1 \subseteq DB-2$ is

not guaranteed — an entity may be able to know something in principle (it is within theoretical reach) without being able to detect it with current instruments. The reverse may also apply in restricted-instrument environments.

Section 15 — Time-War Operations (Extended)

The following operations are inherited from Grok's OAL framework and adopted without modification: Relabel, Remap, Reclock, Reobserve, Resettle, Rewind **(S)**, Branch **(S)**, Merge **(S)**, Rewrite History **(S)**.

The following four operations are Copilot-original additions:

Operation	Label	Description	Hostile When	Observable Residual
Debt Activation	(D)	Issuing a retroactive ruleset change that triggers Temporal Debt across dependent claims. Not inherently hostile — retroactive corrections are often legitimate governance acts.	Done without a TDR; done to manufacture ambiguity; done without disclosure to affected claim holders.	TDR absent or incomplete; affected claims reclassified silently; cascade depth undeclared.
Boundary Push	(D) / (S)	Intentionally moving the Detection Boundary of an adversary inward — reducing what they can detect — without reducing their Knowable frontier. The adversary knows something exists (DB-2) but cannot confirm it (DB-1 shrunk).	Done to prevent an adversary from gathering evidence for a dispute or challenge.	Adversary's DB record shows expanded DB-2 with shrunk DB-1; Observer Network audit shows censorship signatures.

Operation	Label	Description	Hostile When	Observable Residual
Canon Laundering	(S)	Promoting a claim to canon through a CPR that falsifies the evidence package, circumvents the independent review requirement, or omits dissenting opinions. The item is canonically valid (governance status) but evidentially fraudulent.	Always hostile — there is no legitimate reason to falsify a CPR.	CPR evidence package inconsistent with ledger records; missing independent reviewer signature; dissenting opinions absent.
Present Capture	(S)	Taking control of the present-time claim infrastructure for a territory so that all claims referencing "now" within that territory accept the attacker's present-time claims as authoritative. The functional equivalent of controlling the clock for an entire jurisdiction.	Always hostile in an adversarial context — represents total temporal authority capture.	All Observer Network nodes within territory report attacker-controlled times; cross-border observers detect discontinuity; trust anchor registry shows unauthorized substitution.

Section 16 — Research Phases

Phase 0 — Architect Decision Register

Do not implement against silent meanings. Carry forward Grok's 11 open decisions from the prior framework. Add the following 5 Copilot-original decisions that must be resolved before CTIL protocols can be implemented:

115. **Present-time trust anchor:** What entity or system is the declared trust anchor for present-time claims in this lab campaign? (Required for PCP implementation and I22.)
116. **Maximum cascade depth:** What is the maximum declared cascade depth for Temporal Debt tracing? (Required for I26 and TDR completeness.)
117. **Observer Network independence definition:** What constitutes structural independence in an Observer Network — how many trust tiers must be represented, and is shared infrastructure at any tier disqualifying? (Required for I16 and G10.)
118. **Canon Promotion quorum:** Who may issue a Canon Promotion Record, how many OAL contributors constitute a quorum for independent review, and does the promoting authority require unanimous or majority approval?
119. **Sunset condition format:** What is the required format for a sunset condition? It must be falsifiable and specific — "insufficient evidence" is not a valid sunset condition. A template must be agreed before any canon promotion is attempted.

Phase 1 — Ontology, Glossary, Schema

Produce formal JSON-LD or equivalent schemas for: Event Claim Record (inherited), CLR (new — Section 3), SG (new — Section 4), TDR (new — Section 6), Detection Boundary Record (new — Section 7), CPR (new — Section 11), and Observer Network Graph (new — Section 2.2). All schemas must include field-level documentation and a schema version identifier.

Phase 2 — Reference Model

Implement the deterministic graph engine from prior frameworks. Add: CLR state machine (with TLA+ formal specification), Observer Network graph with trust tier verification, Settlement Graph constructor with Conflict Log generation, and Temporal Debt cascade engine with declared depth limit.

Phase 3 — Deterministic Scenario Suites

Suites A-E are inherited from prior frameworks. The following four suites are Copilot-original additions:

- **Suite F — Temporal Debt Cascade:** Issue 3 retroactive ruleset changes with overlapping affected-claim sets. Measure TDR completeness rate, cascade accuracy at declared depth, and open-residual documentation rate.
- **Suite G — Canon Promotion and Demotion:** Promote 5 claims to canon status; then trigger sunset conditions on 2 of them. Measure CPR completeness, promotion traceability, and demotion record accuracy.
- **Suite H — Observer Network Integrity:** Run corroboration probes with Sybil observers, censorship attacks, and single-source repetition injections. Measure ON independence detection rate and Observer Network Audit recall.
- **Suite I — Present-Time Claim Lifecycle:** Inject unbound "now" references at each of the 13 validation gate stages. Measure PCP enforcement rate per gate and identify which gates are PCP-blind.

Phase 4 — Generated Testing and Adversarial Injection

Fuzz the CLR state machine with randomly ordered legal and illegal transitions. Inject all 23 attack vectors from Section 5.3 against reference model implementations. Measure detection rates per vector per oracle. Record all undetected attacks as open residuals — not as test failures.

Phase 5 — Multi-Contributor Challenge

Each OAL contributor runs their framework's validation protocol against a common test corpus. Results are compared and discrepancies recorded as Conversion-and-Dispute Surface points. No contributor's result is authoritative until settled per the Settlement Graph protocol.

Phase 6 — Canon Integration

Items that have passed Phase 3-5 testing with sufficient evidence and independent review may be submitted for canon promotion per Section 11. All promotions require CPRs. All canonical items carry sunset conditions. The program-level canon record is the aggregate of all CPRs — it is not a master timeline.

Section 17 — Open Paradoxes (Extended)

Paradoxes P1-P5 are inherited from Grok's OAL framework and adopted with attribution. The following four paradoxes are Copilot-original additions.

P6 — The Validator's Clock Paradox

Every validation timestamp is itself a temporal claim requiring validation. The chain of "validate the validator" is infinite unless terminated at a declared trust anchor. But the trust anchor itself is an unvalidated assumption — it is trusted by declaration, not by proof. This is not solvable within the system. It is a design constraint. The correct response is to declare trust anchors explicitly, label them as axioms rather than conclusions, and treat their failure as a first-class contingency. A system that does not declare its trust anchors has hidden assumptions that will eventually conflict with each other or with external reality.

P7 — The Settlement Plurality Paradox

Two authorities, each with valid but overlapping jurisdiction, issue contradictory Settlement Graphs over the same scope. Both SGs are formally valid governance artifacts. A claim that is TRUE in SG-A and FALSE in SG-B is simultaneously valid and invalid — not because of an error, but because the governance system produced two legitimate but contradictory outcomes. The lab's correct response is to retain both SGs and flag the affected claim as DISPUTED. But a downstream system that requires a binary answer — a contract that must be enforced or not, a right that must be exercised or denied — cannot be served without a meta-authority or a tiebreaker rule. And any such meta-authority is itself a political act. There is no logical escape from this paradox: the political is irreducible.

P8 — The Detection Boundary Observer Effect

Measuring the Detection Boundary of an entity requires observing what that entity observes — which may itself be outside the measurer's own Detection Boundary. The DB of entity A cannot be fully known from the perspective of entity B without making assumptions about the mapping between A's and B's frames, instruments, and information access. An entity's DB-1 is not directly observable from outside — it can only be inferred from a model of the entity's instrumentation and position. This means

that DB labeling in test design (Invariant I20) is always a model-based approximation, not a direct measurement. The approximation must be declared as such.

P9 — The Temporal Debt Completeness Problem

To know the full scope of Temporal Debt from a retroactive ruleset change, one must already know all claims that were validated under the old ruleset. But the set of all claims in a distributed, adversarial, multi-contributor environment is never fully known: claims may be distributed across jurisdictions, deliberately hidden, or simply unrecorded. The cascade depth limit in a TDR (Invariant I26) is not a solution to this problem — it is an honest acknowledgment of it. Temporal Debt, as measured by any TDR, is always a lower bound on the true debt. The true debt may be unknowable. The lab's correct response is to declare this explicitly and to treat open residuals not as gaps to be closed, but as permanently acknowledged uncertainties.

Section 18 — Verdict and Independent Recommendation

18.1 Assessment of Prior Contributors

Contributor	Strongest Contribution	Identified Gap	Copilot Position
ChatGPT	Richest layer enumeration in the program — the Temporal Stack provides the most comprehensive taxonomy of where temporal claims live and where they can go wrong.	Layers are listed, not queried. The stack enumerates surfaces of vulnerability without providing a query model, a validation gate, or a test protocol.	Adopt the Temporal Stack as a reference checklist and layer taxonomy. Do not treat it as an architecture. Supplement with oracle set and gate protocol before use in validation.
Claude	Best append-only instinct in the program. The three-stamp model (happened/observed/recorded) is the clearest articulation of the occurrence-knowledge gap. The insistence that	The three-stamp model is necessary but incomplete. Ruleset version, frame, uncertainty interval, and dependency cone are absent from the model. A claim can satisfy	Extend the three-stamp model with frame, ruleset, uncertainty, and dependency fields. The model is correct as far as it goes — it does not go far enough for adversarial

Contributor	Strongest Contribution	Identified Gap	Copilot Position
	corrections append and do not overwrite is adopted as design rule D3 (I3).	happened/observed/recorded and still be frame-ambiguous, precision-inflated, or ruleset-orphaned.	environments.
Gemini	The authority-provenance model correctly identifies that the lineage of who made a claim matters, not just what the claim says. Provenance Oracle traces are structurally useful.	"Highest authority wins = truth" conflates governance with validation. An authority can be wrong, captured, or outside its jurisdiction. Authority determines governance outcomes; it does not determine evidentiary correctness.	Retain the authority-provenance model for governance-only applications — determining who has decision rights over a claim. Reject it for validation purposes. Governance and validation are distinct functions.
Perplexity	Most methodologically rigorous prior artifact. TITF's epistemic label system (E/D/S), oracle set, phased research plan, and full test matrix are the most implementation-ready elements in the prior packet. The insistence on naming the asset class and leaving architect decisions explicitly open is the correct posture for a multi-contributor research program.	TITF does not address Observer Network topology, Claim Lifecycle Records, Temporal Debt, or the Present-as-Claim Principle. The oracle set is powerful but does not include CLR, Present-Time, or Temporal Debt oracles.	Adopt TITF's oracle set, phased plan, and test matrix as design rules. Extend with four additional oracles (CLR, Observer Network Audit, Temporal Debt, Present-Time). Adopt the epistemic label system without modification.
Grok OAL	Most operationally rigorous prior artifact. The four unit kinds, primitive object set, integrity principles P1-P8, invariants I1-I15, Time-War operations table, formal modeling suggestions, and adversarial probe	Grok's framework does not model Observer Networks as structured systems, does not specify a Claim Lifecycle Record, does not address Temporal Debt from retroactive ruleset changes, and does not state the	Adopt Grok's primitive objects, invariants I1-I15, Time-War operations, and formal modeling suggestions with extensions declared in this document. Extend the test battery with T11-T30. Credit Grok's

Contributor	Strongest Contribution	Identified Gap	Copilot Position
	design are the most complete operational specification in the prior packet. Grok's verdict on the prior Copilot master timeline contribution is correct.	Present-as-Claim Principle. The test battery (T1-T10) does not cover CLR, PCP, or Temporal Debt failure modes.	Principle P8 for the Detection Boundary insight.

18.2 Copilot's Independent Statement

Temporal integrity is not a property of a timestamp. It is not a property of a clock, an authority, or a settlement. It is a property of a claim's entire lifecycle: the initial assertion, the binding of that assertion to an event, the positioning of that binding within a causal structure, the corroboration of that binding by structurally independent observers, the governance of that corroboration by a jurisdictionally-bounded authority, the settlement of any challenges through an explicitly-labeled Settlement Graph, the tracking of all downstream claims that depend on this one, and the permanent preservation of every state this claim has ever occupied.

A system that does only some of these things is not a temporal integrity system with gaps — it is an incomplete specification waiting to be exploited. The gaps are not omissions; they are attack surfaces. Unvalidated "now" references are attack surfaces. Atomic-witness corroboration models are attack surfaces. Master timelines imposed before settlement are attack surfaces. Ruleset changes issued without Temporal Debt Records are attack surfaces. These are not hypothetical concerns — they are the predictable consequences of underspecified temporal systems.

The five most consequential architectural choices for any temporal system are: whether validations terminate at a declared trust anchor or recurse infinitely; whether retroactive ruleset changes are tracked as Temporal Debt or silently applied; whether Observer Networks are modeled as structured graphs with trust tiers or as atomic witnesses; whether Settlement Graphs are labeled as political acts or presented as measurements; and whether "now" is treated as a validated claim or a background assumption. Getting all five right does not produce a perfect temporal system. It produces a system whose failures are visible, whose assumptions are declared, and whose residuals are preserved for future investigators. That is the appropriate goal for

a research program operating without a universal clock, without a master authority, and without the luxury of assuming good faith from all participants.

That is enough to start the battery. It is not yet canon.

Section 19 — Minimum Deliverables (Extended)

The following deliverables are required before CTIL Phase 3 may begin. Items marked **NEW** are Copilot-original additions to the prior TITF/Grok deliverable list.

- 120. Event Claim Record schema with valid and invalid examples (inherited).
- 121. Versioned Mapping Function schema with validity interval declaration (inherited).
- 122. Observer Network topology specification (inherited, extended with trust tier definitions).
- 123. Rights token instrument schema with enumerated rights (inherited).
- 124. Full oracle set implementation specifications for all 13 oracles (inherited + extended).
- 125. Reference model deterministic graph engine (inherited).
- 126. Temporal Threat Model matrix — complete 22+ attack vectors per Section 5.3 **[NEW]**.
- 127. Claim Lifecycle Record schema with valid example, invalid example, and ZOMBIE detection protocol **[NEW]**.
- 128. Settlement Graph schema with Conflict Log format and malformation criteria **[NEW]**.
- 129. Temporal Debt Record schema with cascade depth declaration and open-residual format **[NEW]**.
- 130. Detection Boundary record format with cutoff timestamp binding and drift log **[NEW]**.

- 131. Canon Promotion Record schema with sunset condition format and demotion record format **[NEW]**.
- 132. CLR state machine formal specification (TLA+ recommended) **[NEW]**.
- 133. Observer Network independence test protocol with Sybil resistance criteria **[NEW]**.
- 134. Present-Time Claim binding validation procedure with all five PCP components **[NEW]**.
- 135. Extended test suite T11–T30 with fixture files and expected oracle responses **[NEW]**.
- 136. Architect Decision Register with all 16 open decisions (11 inherited + 5 Copilot-original) formally recorded and assigned for resolution **[NEW]**.

— End of draft. Not canon. Not a claim on any live system. —

Draft ≠ canon · 29 September 2026 · Copilot independent execution

Copilot Temporal Integrity Lab (CTIL) · OAL Multi-Contributor Temporal Integrity
Research Program

This document is a draft research framework. It does not make claims about any existing physical, legal, computational, or virtual system. All speculative extensions (S) are fictional and must be explicitly stipulated before use.